Chapter 1

Sobolev Spaces

This chapter is devoted to developing function spaces that are used in the variational formulation of differential equations. We begin with a review of Lebesgue integration theory, upon which our notion of "variational" or "weak" derivative rests. Functions with such "generalized" derivatives make up the spaces commonly referred to as Sobolev spaces. We develop only a small fraction of the known theory for these spaces—just enough to establish a foundation for the finite element method.

1.1 Review of Lebesgue Integration Theory

We will now review the basic concepts of Lebesgue integration theory, cf. (Halmos 1991), (Royden 1988) or (Rudin 1987). By "domain" we mean a Lebesgue-measurable (usually either open or closed) subset of $\mathbb{R}^n$ with non-empty interior. We restrict our attention for simplicity to real-valued functions, f, on a given domain, Ω, that are Lebesgue measurable; by

$$\int_\Omega f(x)\,dx$$

we denote the Lebesgue integral of f (dx denotes Lebesgue measure). For $1 \le p < \infty$, let

$$\|f\|_{L^p(\Omega)} := \left(\int_\Omega |f(x)|^p\,dx\right)^{1/p},$$

and for the case $p = \infty$ set

$$\|f\|_{L^\infty(\Omega)} := \text{ess sup}\,\{|f(x)| \;:\; x \in \Omega\}\,.$$

In either case, we define the *Lebesgue spaces*

$$L^p(\Omega) := \{f \;:\; \|f\|_{L^p(\Omega)} < \infty\}. \tag{1.1.1}$$

To avoid trivial differences between functions, we identify two functions, f and g, that satisfy $\|f - g\|_{L^p(\Omega)} = 0$. For example, take $n = 1$, $\Omega = [-1, 1]$ and

(1.1.2) $$f(x) := \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \quad \text{and} \quad g(x) := \begin{cases} 1 & x > 0 \\ 0 & x \leq 0. \end{cases}$$

Since f and g differ only on a set of measure zero (one point, in this case), we view them as representing the same function. With a small ambiguity of notation, we then think of $L^p(\Omega)$ as a set of *equivalence classes* of functions with respect to this identification. There are some famous (and useful) inequalities that hold for the functionals defined above:

(1.1.3) **Minkowski's Inequality** *For $1 \leq p \leq \infty$ and $f, g \in L^p(\Omega)$, we have*

$$\|f + g\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)} + \|g\|_{L^p(\Omega)} \,.$$

(1.1.4) **Hölder's Inequality** *For $1 \leq p, q \leq \infty$ such that $1 = 1/p + 1/q$, if $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$, then $f\,g \in L^1(\Omega)$ and*

$$\|f\,g\|_{L^1(\Omega)} \leq \|f\|_{L^p(\Omega)} \, \|g\|_{L^q(\Omega)} \,.$$

(1.1.5) **Schwarz' Inequality** *This is simply Hölder's inequality in the special case $p = q = 2$, viz. if $f, g \in L^2(\Omega)$ then $f\,g \in L^1(\Omega)$ and*

$$\int_\Omega |f(x)g(x)| \, dx \leq \|f\|_{L^2(\Omega)} \, \|g\|_{L^2(\Omega)} \,.$$

In view of Minkowski's inequality and the definitions of $\|\cdot\|_{L^p(\Omega)}$, the space $L^p(\Omega)$ is closed under linear combinations, i.e., it is a *linear* (or *vector*) space. Moreover, the functionals $\|\cdot\|_{L^p(\Omega)}$ have properties that classify them as *norms*.

(1.1.6) Definition. *Given a linear (vector) space V, a* **norm**, $\|\cdot\|$, *is a function on V with values in the non-negative reals having the following properties:*

i) $\|v\| \geq 0 \quad \forall\, v \in V$
$\|v\| = 0 \iff v = 0$

ii) $\|c \cdot v\| = |c| \cdot \|v\| \quad \forall c \in \mathbb{R}, v \in V,$ and

iii) $\|v + w\| \leq \|v\| + \|w\| \quad \forall\, v, w \in V$ (**the triangle inequality**).

A norm, $\|\cdot\|$, can be used to define a notion of distance, or *metric*, $d(v, w) = \|v - w\|$ for points $v, w \in V$. A vector space endowed with the topology induced by this metric is called a *normed linear space*. Recall that a metric space, V, is called *complete* if every *Cauchy* sequence $\{v_j\}$ of elements of V has a limit $v \in V$. For a normed linear space, a Cauchy sequence is one such that $\|v_j - v_k\| \to 0$ as $j, k \to \infty$, and completeness means that $\|v - v_j\| \to 0$ as $j \to \infty$. The following definition encapsulates some

key features of linear spaces of infinite dimensions needed for theoretical development.

(1.1.7) Definition. *A normed linear space* $(V, \|\cdot\|)$ *is called a* **Banach space** *if it is complete with respect to the metric induced by the norm,* $\|\cdot\|$.

(1.1.8) Theorem. *For* $1 \le p \le \infty$, $L^p(\Omega)$ *is a Banach space.*

This theorem (whose proof may be found in the references at the beginning of this section) is a cornerstone of Lebesgue integration theory. Note that it incorporates both Minkowski's inequality and a limit theorem for the Lebesgue integral; that is, if $f_j \to f$ in $L^p(\Omega)$ then (cf. exercise 1.x.3)

$$\int_\Omega |f_j(x)|^p \, dx \to \int_\Omega |f(x)|^p \, dx \quad \text{as} \quad j \to \infty. \tag{1.1.9}$$

However, Hölder's inequality and more subtle limit theorems are not reflected in this characterization of the Lebesgue spaces.

A key reason that the Lebesgue integral is preferred over the Riemann integral is the aspect of "completeness" that it enjoys, i.e., that appropriate limits of integrable functions are integrable, a property that the Riemann integral does not have. For example, we can easily evaluate an improper integral to determine that the function $\log\, x$ has a finite integral on any finite interval of the form $[0, a]$. Correspondingly, if $\{r_n \,:\, n = 1, 2, \ldots\}$ is dense in the interval $[0, 1]$, then the functions

$$f_j(x) := \sum_{n=1}^{j} 2^{-n} \log\, |x - r_n| \tag{1.1.10}$$

all have improper integrals, and one easily sees that

$$\left| \int_0^1 f_j(x) \, dx \right| \le 2 \int_0^1 |\log\, x| \;\; dx. \tag{1.1.11}$$

Therefore, the "limit" function

$$f(x) := \sum_{n=1}^{\infty} 2^{-n} \log\, |x - r_n| \tag{1.1.12}$$

should have a finite "integral" on $[0, 1]$, again satisfying

$$\left| \int_0^1 f(x) \, dx \right| \le 2 \int_0^1 |\log\, x| \;\; dx. \tag{1.1.13}$$

However, f is infinite at some point in any open sub-interval of $[0, 1]$ and so it is not Riemann integrable on any sub-interval of $[0, 1]$. Thus, it is not possible, even via "improper" Riemann integrals, to determine if f has a finite integral. On the other hand, one can show (see exercise 1.x.6) that it is Lebesgue integrable and that (1.1.13) holds.

1.2 Generalized (Weak) Derivatives

There are several definitions of derivative that are useful in different situations. The "calculus" definition, viz.

$$u'(x) = \lim_{h \to 0} \frac{u(x+h) - u(x)}{h},$$

is a "local" definition, involving information about the function u only near the point x. The variational formulation developed in Chapter 0 takes a more global view, because pointwise values of derivatives are not needed; only derivatives that can be interpreted as functions in the Lebesgue space $L^2(\Omega)$ occur. In the previous section, we have seen that pointwise values of functions in Lebesgue spaces are irrelevant (cf. (1.1.2)); a function in one of these spaces is determined only by its global behavior. Thus, it is natural to develop a global notion of *derivative* more suited to the Lebesgue spaces. We do so using a "duality" technique, defining derivatives for a class of not-so-smooth functions (see Definition 1.2.3) by comparing them with very-very-smooth functions (introduced in Definition 1.2.1).

First, let us introduce some short-hand notation for (calculus) partial derivatives, the *multi-index* notation. A multi-index, α, is an n-tuple of non-negative integers, α_i. The length of α is given by

$$|\alpha| := \sum_{i=1}^{n} \alpha_i.$$

For $\phi \in C^\infty$, denote by

$$D^\alpha \phi, \quad D_x^\alpha \phi, \quad \left(\frac{\partial}{\partial x}\right)^\alpha \phi, \quad \phi^{(\alpha)}, \quad \text{and} \quad \partial_x^\alpha \phi$$

the usual (pointwise) partial derivative

$$\left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n} \phi.$$

Given a vector $(x_1, \ldots, x_n)$, we define $x^\alpha := x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n}$. Note that if x is replaced formally by the symbol $\frac{\partial}{\partial x} := \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n}\right)$, then this definition of x^α is consistent with the previous definition of $\left(\frac{\partial}{\partial x}\right)^\alpha$. Note that the *order* of this derivative is given by $|\alpha|$.

Next, let us introduce the concept of the *support* of a function defined on some domain in $\mathbb{R}^n$. For a continuous function, u, this is the closure of the (open) set $\{x \,:\, u(x) \neq 0\}$. If this is a compact set (i.e., if it is bounded) and it is a subset of the *interior* of a set, Ω, then u is said to have "compact support" with respect to Ω. (Outside the support of a function, it is natural to define it to be zero, thus extending it to be defined on all of $\mathbb{R}^n$.) When

Ω is a bounded set, it is equivalent to say that u vanishes in a neighborhood of $\partial\Omega$.

(1.2.1) Definition. *Let Ω be a domain in $\mathbb{R}^n$. Denote by $\mathcal{D}(\Omega)$ or $C_0^\infty(\Omega)$ the set of $C^\infty(\Omega)$ functions with compact support in Ω.*

Before proceeding any further, it would be wise to verify that we have not just introduced a vacuous definition, which we do in the following:

(1.2.2) Example. Define

$$\phi(x) := \begin{cases} e^{1/(|x|^2-1)} & |x| < 1 \\ 0 & |x| \geq 1 \ . \end{cases}$$

We claim that, for any multi-index α, $\phi^{(\alpha)}(x) = P_\alpha(x)\phi(x)/(1-|x|^2)^{|\alpha|}$ for some polynomial P_α, as we now show. For $|x| < 1$, we can differentiate and determine, inductively in α, that $\phi^{(\alpha)}(x) = P_\alpha(x)e^{-t}t^{|\alpha|}$ for some polynomial P_α, where $t = 1/(1-|x|^2)$. Further, $\phi^{(\alpha)}(x) = 0$ for $|x| > 1$. Thus, the formula above for $\phi^{(\alpha)}$ is verified in the case $|x| \neq 1$. Since the exponential increases faster than any finite power, $\phi^{(\alpha)}(x)/\left(1-|x|^2\right)^k = P_\alpha(x)t^{|\alpha|+k}/e^t \to 0$ as $|x| \to 1$ (i.e., as $t \to \infty$) for any integer k. Applying (inductively) these facts with $k = 0$ shows that $\phi^{(\alpha)}$ is continuous at $|x| = 1$, and using $k = 1$ shows it is also differentiable there, and has derivative zero. Thus, the claimed formula holds for all x. Moreover, we also see from the argument that $\phi^{(\alpha)}$ is bounded and continuous for all α. Thus, $\phi \in \mathcal{D}(\Omega)$ for any open set Ω containing the closed unit ball. By scaling variables appropriately, we see that $\mathcal{D}(\Omega) \neq \emptyset$ for any Ω with non-empty interior.

We now use the space $\mathcal{D}$ to extend the notion of pointwise derivative to a class of functions larger than C^∞. For simplicity, we restrict our notion of derivatives to the following space of functions (see (Schwartz 1957) for a more general definition).

(1.2.3) Definition. *Given a domain Ω, the set of* **locally integrable** *functions is denoted by*

$$L^1_{loc}(\Omega) := \left\{ f \ : \ f \in L^1(K) \quad \forall \text{ compact } K \subset \text{interior}\, \Omega \right\}.$$

Functions in $L^1_{loc}(\Omega)$ can behave arbitrarily badly near the boundary, e.g., the function $e^{e^{1/\mathrm{dist}(x,\partial\Omega)}} \in L^1_{loc}(\Omega)$, although this aspect is somewhat tangential to our use of the space. One notational convenience is that $L^1_{loc}(\Omega)$ contains all of $C^0(\Omega)$, without growth restrictions. Finally, we come to our new definition of derivative.

(1.2.4) Definition. *We say that a given function $f \in L^1_{loc}(\Omega)$ has a* **weak derivative**, $D^\alpha_w f$, *provided there exists a function $g \in L^1_{loc}(\Omega)$ such that*

$$\int_\Omega g(x)\phi(x)\,dx \;=\; (-1)^{|\alpha|}\int_\Omega f(x)\phi^{(\alpha)}(x)\,dx \quad \forall \phi \in \mathcal{D}(\Omega).$$

If such a g exists, we define $D^\alpha_w f = g$.

(1.2.5) Example. Take $n = 1$, $\Omega = [-1, 1]$, and $f(x) = 1 - |x|$. We claim that $D^1_w f$ exists and is given by

$$g(x) := \begin{cases} 1 & x < 0 \\ -1 & x > 0. \end{cases}$$

To see this, we break the interval $[-1, 1]$ into the two parts in which f is smooth, and we integrate by parts. Let $\phi \in \mathcal{D}(\Omega)$. Then

$$\begin{aligned}
\int_{-1}^1 f(x)\phi'(x)\,dx \;&=\; \int_{-1}^0 f(x)\phi'(x)\,dx + \int_0^1 f(x)\phi'(x)\,dx \\
&=\; -\int_{-1}^0 (+1)\phi(x)\,dx + f\phi|_{-1}^0 - \int_0^1 (-1)\phi(x)\,dx + f\phi|_0^1 \\
&=\; -\int_{-1}^1 g(x)\phi(x)\,dx + (f\phi)(0-) - (f\phi)(0+) \\
&=\; -\int_{-1}^1 g(x)\phi(x)\,dx
\end{aligned}$$

because f is continuous at 0. One may check (cf. exercise 1.x.10) that $D^j_w f$ does not exist for $j > 1$.

One can see that, roughly speaking, the new definition of derivative is the same as the old one wherever the function being differentiated is regular enough. In particular, continuity of f in the example was enough to insure existence of a first-order weak derivative, but not second-order. This phenomenon depends on the dimension n as well, precluding a simple characterization of the relation between the calculus and weak derivatives, as the following example shows.

(1.2.6) Example. Let ρ be a smooth function defined for $0 < r \le 1$ satisfying

$$\int_0^1 |\rho'(r)|\, r^{n-1}\,dr < \infty.$$

Define f on $\Omega = \{x \in \mathbb{R}^n \;:\; |x| < 1\}$ via $f(x) = \rho(|x|)$. Then $D^\alpha_w f$ exists for all $|\alpha| = 1$ and is given by

$$g(x) \;=\; \rho'(|x|)x^\alpha/|x|.$$

The verification of this is left to the reader in exercise 1.x.12.

This example shows that the relationship between the calculus and weak derivatives depends on dimension. That is, whether a function such as $|x|^r$ has a weak derivative depends not only on r but also on n (cf. exercise 1.x.13). However, the following fact (whose proof is left as an exercise) shows that the latter is a generalization of the former.

(1.2.7) Proposition. *Let α be arbitrary and let $\psi \in C^{|\alpha|}(\Omega)$. Then the weak derivative $D_w^\alpha \psi$ exists and is given by $D^\alpha \psi$.*

As a consequence of this proposition, we ignore the differences in definition of D and D_w from now on. That is, differentiation symbols will refer to weak derivatives in general, but we will also use classical properties of derivatives of smooth functions as appropriate.

1.3 Sobolev Norms and Associated Spaces

Using the notion of weak derivative, we can generalize the Lebesgue norms and spaces to include derivatives.

(1.3.1) Definition. *Let k be a non-negative integer, and let $f \in L^1_{loc}(\Omega)$. Suppose that the weak derivatives $D_w^\alpha f$ exist for all $|\alpha| \le k$. Define the* **Sobolev norm**

$$\|f\|_{W_p^k(\Omega)} := \left(\sum_{|\alpha| \le k} \|D_w^\alpha f\|^p_{L^p(\Omega)} \right)^{1/p}$$

in the case $1 \le p < \infty$, and in the case $p = \infty$

$$\|f\|_{W_\infty^k(\Omega)} := \max_{|\alpha| \le k} \|D_w^\alpha f\|_{L^\infty(\Omega)} .$$

In either case, we define the **Sobolev spaces** *via*

$$W_p^k(\Omega) := \left\{ f \in L^1_{loc}(\Omega) \ : \ \|f\|_{W_p^k(\Omega)} < \infty \right\} .$$

The Sobolev spaces can be related in special cases to other spaces. For example, recall the *Lipschitz norm*

$$\|f\|_{Lip(\Omega)} \ = \ \|f\|_{L^\infty(\Omega)} + \sup \left\{ \frac{|f(x) - f(y)|}{|x - y|} \ : \ x, y \in \Omega; x \ne y \right\},$$

and the corresponding space of *Lipschitz functions*

$$Lip(\Omega) \;=\; \Big\{ f \in L^\infty(\Omega) \;:\; \|f\|_{Lip(\Omega)} < \infty \Big\}.$$

Then for all dimensions n, we have $Lip(\Omega) = W^1_\infty(\Omega)$ with equivalent norms, at least under certain conditions on the domain Ω (cf. exercises 1.x.15 and 1.x.14). (Two norms, $\|\cdot\|_1$ and $\|\cdot\|_2$, on a linear space V are said to be *equivalent* provided there is a positive constant $C < \infty$ such that $\|v\|_1/C \le \|v\|_2 \le C\|v\|_1 \quad \forall v \in V$.) Moreover, for $k > 1$

$$W^k_\infty(\Omega) \;=\; \Big\{ f \in C^{k-1}(\Omega) \;:\; f^{(\alpha)} \in Lip(\Omega)\, \forall\, |\alpha| \le k-1 \Big\}.$$

In one dimension ($n = 1$), the space $W^1_1(\Omega)$ can be characterized as the set of *absolutely continuous* functions on an interval Ω (cf. (Hartman & Mikusinski 1961) and exercises 1.x.17 and 1.x.22).

It is easy to see that $\|\cdot\|_{W^k_p(\Omega)}$ is a norm. Thus, $W^k_p(\Omega)$ is by definition a normed linear space. The following theorem shows that it is complete.

(1.3.2) Theorem. *The Sobolev space $W^k_p(\Omega)$ is a Banach space.*

Proof. Let $\{v_j\}$ be a Cauchy sequence with respect to the norm $\|\cdot\|_{W^k_p(\Omega)}$. Since the $\|\cdot\|_{W^k_p(\Omega)}$ norm is just a combination of $\|\cdot\|_{L^p(\Omega)}$ norms of weak derivatives, it follows that, for all $|\alpha| \le k$, $\{D^\alpha_w v_j\}$ is a Cauchy sequence with respect to the norm $\|\cdot\|_{L^p(\Omega)}$. Thus, Theorem 1.1.8 implies the existence of $v^\alpha \in L^p(\Omega)$ such that $\|D^\alpha_w v_j - v^\alpha\|_{L^p(\Omega)} \to 0$ as $j \to \infty$. In particular, $v_j \to v^{(0,\dots,0)} =: v$ in $L^p(\Omega)$. What remains to check is that $D^\alpha_w v$ exists and is equal to v^α.

First, note that if $w_j \to w$ in $L^p(\Omega)$, then for all $\phi \in \mathcal{D}(\Omega)$

$$\int_\Omega w_j(x)\phi(x)\,dx \to \int_\Omega w(x)\phi(x)\,dx. \tag{1.3.3}$$

This follows from (1.1.9) and Hölder's inequality:

$$\|w_j\phi - w\phi\|_{L^p(\Omega)} \le \|w_j - w\|_{L^p(\Omega)}\,\|\phi\|_{L^\infty(\Omega)} \to 0 \quad \text{as} \quad j \to \infty.$$

To show that $D^\alpha_w v = v^\alpha$, we must show that

$$\int_\Omega v^\alpha \phi\,dx \;=\; (-1)^{|\alpha|}\int_\Omega v\phi^{(\alpha)}\,dx \quad \forall \phi \in \mathcal{D}(\Omega).$$

This follows from the definition of the weak derivative, $D^\alpha_w v_j$, and two applications of (1.3.3):

$$\begin{aligned}\int_\Omega v^\alpha \phi\,dx \;&=\; \lim_{j\to\infty}\int_\Omega (D^\alpha_w v_j)\,\phi\,dx \\ &=\; \lim_{j\to\infty}(-1)^{|\alpha|}\int_\Omega v_j\phi^{(\alpha)}\,dx \;=\; (-1)^{|\alpha|}\int_\Omega v\phi^{(\alpha)}\,dx. \qquad \square\end{aligned}$$

There is another potential definition of Sobolev space that could be made. Let $H_p^k(\Omega)$ denote the closure of $C^k(\Omega)$ with respect to the Sobolev norm $\|\cdot\|_{W_p^k(\Omega)}$. In the case $p = \infty$, we have $H_\infty^k = C^k$, and this is not the same as $W_\infty^k(\Omega)$. Indeed, we have already identified the latter as being related to certain Lipschitz spaces. However, for $1 \le p < \infty$, it turns out that $H_p^k(\Omega) = W_p^k(\Omega)$. The following result was proved in a paper (Meyers & Serrin 1964) that is celebrated both for the importance of the result and the brevity of its title.

(1.3.4) Theorem. *Let Ω be any open set. Then $C^\infty(\Omega) \cap W_p^k(\Omega)$ is dense in $W_p^k(\Omega)$ for $p < \infty$.*

(1.3.5) *Remark.* This result should be contrasted with another kind of density result, namely, the density of $C^\infty(\overline{\Omega})$ in Sobolev spaces. The latter cannot happen whenever part of the domain lies on both sides of part of its boundary, as occurs with a slit domain that is frequently used to model crack propagation problems (cf. exercise 1.x.26). In order for this stronger density result to hold, some sort of regularity condition must also hold. For example, it is known (cf. Adams 1975) to be valid if Ω satisfies the *segment* condition, i.e., if for all $x \in \partial\Omega$ there is an open ball B_x containing x and a non-trivial vector n_x such that for all $z \in \overline{\Omega} \cap B_x$ the segment $\{z + tn_x \,:\, 0 < t < 1\} \subset \Omega$. The vector n_x plays the role of an inward-directed normal to $\partial\Omega$ at x. Many of the domains considered here satisfy this condition (cf. exercise 1.x.25).

(1.3.6) *Remark.* The validity of the theorem can be seen in the case $\Omega = \mathbb{R}^n$ as follows. (For a more general case, see exercise 1.x.19.) From Example 1.2.2, we see that there exist non-negative $\phi \in \mathcal{D}(\Omega)$, and so by normalizing appropriately, we can assume that there is such a $\phi \in \mathcal{D}(\Omega)$ further satisfying $\int_\Omega \phi(x)\,dx = 1$. For any $f \in L_{loc}^1(\Omega)$ and $\epsilon > 0$, define

$$f_\epsilon(x) \;=\; \epsilon^{-n} \int_\Omega f(y)\phi\left((x-y)/\epsilon\right) dy,$$

that is, we have $f_\epsilon = f * \phi^\epsilon$ where $\phi^\epsilon(y) := \epsilon^{-n}\phi(y/\epsilon) \quad \forall y \in \Omega$ and "$*$" denotes convolution. From the dominated convergence theorem, it is easy to see that f_ϵ is C^∞ and that $f \in L^p(\Omega)$ implies $f_\epsilon \to f$ in $L^p(\Omega)$. Moreover, if $f \in W_p^k(\Omega)$ then the same can be said of the derivatives of f_ϵ and f, by differentiating under the integral sign and using the definition of weak derivative, thus proving the result. The function f_ϵ is called a "mollification" of f.

For technical reasons it is useful to introduce the following notation for the Sobolev *semi-norms*.

(1.3.7) Definition. *For k a non-negative integer and $f \in W_p^k(\Omega)$, let*

$$|f|_{W_p^k(\Omega)} = \left(\sum_{|\alpha|=k} \|D_w^\alpha f\|_{L^p(\Omega)}^p \right)^{1/p}$$

in the case $1 \le p < \infty$, and in the case $p = \infty$

$$|f|_{W_\infty^k(\Omega)} = \max_{|\alpha|=k} \|D_w^\alpha f\|_{L^\infty(\Omega)} .$$

1.4 Inclusion Relations and Sobolev's Inequality

Given the number of indices defining Sobolev spaces, it is natural to hope that there are inclusion relations to provide some sort of ordering among them. Using the Definition 1.3.1 and exercise 1.x.1, it is easy to derive the following propositions.

(1.4.1) Proposition. *Suppose that Ω is any domain, k and m are non-negative integers satisfying $k \le m$, and p is any real number satisfying $1 \le p \le \infty$. Then $W_p^m(\Omega) \subset W_p^k(\Omega)$.*

(1.4.2) Proposition. *Suppose that Ω is a bounded domain, k is a non-negative integer, and p and q are real numbers satisfying $1 \le p \le q \le \infty$. Then $W_q^k(\Omega) \subset W_p^k(\Omega)$.*

However, there are more subtle relations among the Sobolev spaces. For example, there are cases when $k < m$ and $p > q$ and $W_q^m(\Omega) \subset W_p^k(\Omega)$. The existence of Sobolev derivatives imply a stronger integrability condition of a function. To set the stage, let us consider an example to give us guidance as to possible relations among k, m, p, and q for such a result to hold.

(1.4.3) Example. Let $n \ge 2$, let $\Omega = \{x \in \mathbb{R}^n : |x| < 1/2\}$ and consider the function $f(x) = \log \big| \log |x| \big|$. From Example 1.2.6 (and exercise 1.x.12), we see that f has first-order weak derivatives

$$D^\alpha f(x) = x^\alpha / \left(|x|^2 \log |x| \right)$$

($|\alpha| = 1$). From exercise 1.x.5, we see that $D^\alpha f \in L^p(\Omega)$ provided $p \le n$. For example,

$$|D^\alpha f(x)|^n \le \rho(|x|) := 1/ \left(|x|^n \big| \log |x| \big|^n \right)$$